

NGC 2440 nucleus (HD 62166) **268**
NGC 2447 290
NGC 2451 409, 437
NGC 2477 409, 409, 437
NGC 2516 411, 437
NGC 2547 410
NGC 2736 (Vela Supernova) **269**, 410
NGC 2755 302
NGC 2787 304
NGC 2841 73, 303
NGC 2997 396, 396
NGC 3031 (Bode's Galaxy) **314**, 360, 360
NGC 3034 (Cigar Galaxy) 73, 305, **314**, 360
NGC 3079 73
NGC 3114 411
NGC 3115 (Spindle Galaxy) **317**, 396, 396
NGC 3132 (Eight-Burst Nebula) 254–255, 410, 443
NGC 3195 423
NGC 3201 **294**
NGC 3242 (Ghost of Jupiter) 394, 394
NGC 3309 332
NGC 3311 332, 332
NGC 3312 332
NGC 3314 332
NGC 3372 *see* Carina Nebula
NGC 3532 411, 411
NGC 3603 234
NGC 3628 377
NGC 3766 398
NGC 3918 (Blue Planetary) 398
NGC 4038 and 4039 (Antennae Galaxies) 37, 309, **317**, 318, 397, 397
NGC 4261 **323**
NGC 4449 305
NGC 4472 307
NGC 4486 323
NGC 4526 283
NGC 4565 376, 376
NGC 4590 295
NGC 4594 (Sombrero Galaxy) **316**, 378, 378
NGC 4621 307
NGC 4622 302
NGC 4649 317
NGC 4650A 305
NGC 4676 (the Mice) 318
NGC 4755 (Jewel Box) **294**, 412, 412, 449, 455, 461
NGC 4826 (Black Eye Galaxy) **314**, 376
NGC 4833 **295**, 413
NGC 4881 332
NGC 4889 326, 332
NGC 5128 *see* Centaurus A
NGC 5139 *see* Omega Centauri
NGC 5194 and NGC 5195 (Whirlpool Galaxy) 14, 302, **315**, 362, 454, 460
NGC 5457 (Pinwheel Galaxy) **316**, 360, 454, 460
NGC 5460 398
NGC 5548 **323**
NGC 5822 399, 399
NGC 5866 317
NGC 5907 308
NGC 6025 414
NGC 6027 and NGC 6027A–C (Seyfert's Sextet) 308, **329**
NGC 6041A 333

NGC 6050 308, 333
NGC 6087 414, 414
NGC 6121 294
NGC 6128 295
NGC 6171 295
NGC 6193 415
NGC 6231 402, 461
NGC 6302 (Bug Nebula) **260–261**
NGC 6397 415, 415
NGC 6402 295
NGC 6405 *see* Butterfly Cluster
NGC 6514 246, 246
NGC 6523 243
NGC 6530 243, 400, 467
NGC 6541 415
NGC 6543 (Cat's Eye Nebula) **258**, 355, 355
NGC 6633 381, 466
NGC 6744 424, 424
NGC 6751 25, 255
NGC 6752 424, 424
NGC 6782 **318**
NGC 6822 (Barnard's Galaxy) 328
NGC 6826 367
NGC 6888 (Crescent Nebula) **259**
NGC 6960/95 (Cygnus Loop) 37, 228, **269**
NGC 6992 367
NGC 7000 (North America Nebula) 367, 367, 478
NGC 7009 (Saturn Nebula) 255, 387, 387, 479
NGC 7078 295
NGC 7293 (Helix Nebula) **257**, 387, 387, 479
NGC 7320 332, 332
NGC 7479 302
NGC 7654 290
NGC 7662 (Blue Snowball) 368, 368, 484
NGC 7742 (Fried Egg Galaxy) **323**
Niagara Falls (Earth) 134
Nicholson Regio (Ganymede) 186
nickel Earth 126
meteorites 222, 223
nitrogen Bug Nebula 260
Earth's atmosphere 128
main-sequence stars 250
meteorites 223
planetary nebulae 255
Pluto 208–209, 208–209
Titan 196
Wolf-Rayet stars 255
Nix 209, 209
Noachis Terra Crater 151
Noah 408
noctilucent clouds 75, 75, 460
Noctis Labyrinthus (Mars) 154–155, 158
Norma (the Set Square) **414**, 461
Ant Nebula **259**
Epsilon (ε) Normae 414
Gamma-1 (γ¹) Normae 414
Gamma-2 (γ²) Normae 414
Iota (ι) Normae 414
Norma Cluster 336
Norse mythology, Polaris 279
North America Nebula (NGC 7000) 367, 367, 478
north celestial pole 62
North Polar Region (Mars) 153, **161**

North Polar sky 348
Northern Coalsack 367
Northern Cross *see* Cygnus
Northern Crown *see* Corona Borealis
Northern Lights 107
novae **282**
binocular astronomy 80
Nova Cygni 1992 282, **287**
RS Ophiuchi **287**
T Coronae Borealis (Blaze Star) **286**
November sky guide **490–495**
Nu (ν) Boötis 363
Nu (ν) Coronae Borealis 379
Nu (ν) Draconis 355
Nu (ν) Scorpii 402
nuclear fission 31
nuclear fusion 31, 31
inside stars 232, 234, 234
main-sequence stars 250
protostars 239
star formation 238
Sun 104
nucleons 30
nucleus, atom 28, 29
after Big Bang 50
forces 30, 30
NuSTAR mission 95
O
O stars 233, 255
O3 stars 247
OB stars 241
Oberon 201, **203**
observable universe 23
observatories **90–95**, 90–95
see also individual named observatories, telescopes
occultation 69, 253
oceans (Earth) 129, 129
tides 138, 138
Octans (the Octant) **425**
Delta (δ) Octantis 425
Sigma (σ) Octantis 425
Octant *see* Octans
October sky guide **484–489**
Odysseus Crater (Tethys) 194
OH231.8+4.2 (Calabash Nebula) 262
oil reserves, on Earth 135
Okmok volcano (Earth) 131
Olber's paradox 51
Olbers, Heinrich 175
old stars **254–265**
globular clusters 289
Olympus Mons (Mars) 152, 152, 156, 156, **157**
OMC-1 241
Omega Centauri (NGC 5139) 288, 289, **294**, 398
binocular astronomy 81
density 289
in monthly sky guides 418, 449, 455, 461
Omega Nebula (M17) 240, 400, 401
Omega (ω) Scorpii 402
Omicron (ο) Ceti (Mira) **285**, 389, 490, 491, 491
Omicron (ο) Cygni 366
Omicron (ο) Draconis 355
Omicron (ο) Eridani **276**, 406, 406
Omicron (ο) Velorum 410

On the Revolution of the Heavenly Spheres 69
Ontario, Lake (Earth) 134
Oort, Jan Hendrik **213**
Oort Cloud **210–213**
open clusters **288**, 290
Beehive Cluster **290**
Butterfly Cluster **290**
evolution 289, 289
Hyades **290**
Jewel Box (Kappa (κ) Crucis) **294**
M52 **290**
M93 **290**
Pleiades **291**
open universe 59, 59
Ophelia 201, **202**
Ophir Chasma (Mars) 159, 159
Ophiuchus (the Serpent Holder) 380, **381**
Barnard's Star 70, 381
Cygnus Rift 367, 472
Kepler's Star **273**, 381
M12 **295**, 381, 466
M14 **295**
M107 **295**
in monthly sky guides 466, 472
Rho (ρ) Ophiuchi 381, 381
RS Ophiuchi **287**
Twin Jet Nebula **257**
Zeta (ζ) Ophiuchi 268
36 Ophiuchi 381
70 Ophiuchi 381
Öpik, Ernst 213
Opportunity rover, on Mars 164, 166–167
opposition, planets 68
optical telescopes 37, 37, 82–87
orange stars Alpha (α) Centauri (Rigil Kentaurus) **252**, 274
61 Cygni **252**
orange-red stars Proxima Centauri **252**
Orbiting Astronomical Observatory 94
orbits asteroids 170, 170–171
comets 214, 214
elliptical galaxies 304
globular clusters 289
Jupiter 178, 178
Kuiper Belt objects 208, 210–212
Lagrange points 194
Mars 150, 150
Mercury 43, 110, 110
Moon 136
multiple stars 274, 274
Neptune 204, 204, 211
Pluto 208, 211
Saturn 188, 188
Sedna 213, 213
shapes of 39, 39
solar system **102–103**
space-time 42–43, 42–43
spiral galaxies 302
synchronous rotation 136
Uranus 200, 200, 211
Venus 116, 116
Orion (the Hunter) 72, **390–391**
Alnilam (Epsilon (ε) Orionis) 232
Alnitak (Zeta (ζ) Orionis) 232, 390, 391, 391
Bellatrix (Gamma (γ) Orionis) 71

Orion *cont.*
Delta (δ) Orionis 390
Ghost Nebula 89
Horsehead Nebula **240**, 241, 391, 391
Iota (ι) Orionis 391
mythology 390, 402
in monthly sky guides 431, 436, 437, 442, 443, 496, 497
naked-eye astronomy 77, 77
Orion's belt 72, 390, 431
Sigma (σ) Orionis 240, **281**, 390, 391
star colors 70–71
see also Betelgeuse (Alpha (α) Orionis); Rigel (Beta (β) Orionis)
Trapezium (Theta (θ) Orionis) 241, 241, 275, 276, **281**, 391, 391
42 Orionis 391
45 Orionis 391
Orion Arm, Milky Way 227, 229, 229
Orion Nebula (M42, NGC 1976) 14–15, 36, **241**, 310, 390, 391, 391
binocular astronomy 81, 81
bow shock 20–21
in monthly sky guides 430, 431, 431
Theta (θ) Orionis 281
young stars 55
Orionid meteor shower 390, 484
Orpheus, mythology 365
Orthosie 181
OSIRIS-REx mission 176
Ovda Regio (Venus) **123**
Owl Nebula 360, 360
oxygen Earth's atmosphere 126, 126
extrasolar planets 299
formation 29, 55
main-sequence stars 250
Mercury 111, 111
meteorites 223
planetary nebulae 255
silicates 24
supergiants 254
Type I supernovae 283
Wolf-Rayet stars 255
Ozza Mons (Venus) 122
P
Paaliaq 191
Pacific Ocean (Earth) 8
Pacific Ring of Fire (Earth) **131**
Painter's Easel *see* Pictor
Palermo Circle 175
Palisa, Johann 173
Pallas 175
Pallene 190, **192**
Palomar Observatory 90
Palus Putredinis (Moon) 139
Pan (Amalthea) 182
Pan (Saturn's moon) 190, 403
Pandia 180
Pandora 190, 191
Papin, Denis 396
parabolas, orbits 39
parallax shift 70, 70
Paranal Observatory 36, 90, 209
parhelia 74